

Limits – 7

Means, inequalities and contractions

1 сентября 2026 года

Theoretical minimum.

Def. The power mean of order p of positive numbers $a_1, \dots, a_n$ is $M_p = \left(\frac{1}{n} \sum a_i^p\right)^{1/p}$, with M_0 being the geometric mean. The function $p \mapsto M_p$ is non-decreasing, and all classical inequalities between means follow from this.

Theorem 1 (Toeplitz and Kronecker). If $t_{n,k} \geq 0$, $\sum_{k=1}^n t_{n,k} = 1$ and $t_{n,k} \rightarrow 0$ for each fixed k , then $\sum_k t_{n,k} x_k \rightarrow \lim x_n$. If $b_n \uparrow \infty$ and the series $\sum \frac{x_n}{b_n}$ converges, then $\frac{1}{b_n} \sum_{k \leq n} x_k \rightarrow 0$ (Kronecker's lemma).

Theorem 2 (contraction). If $|x_{n+1} - L| \leq q|x_n - L|$ with $q < 1$, then $x_n \rightarrow L$ geometrically. The weaker condition $|x_{n+1} - x_n| \leq q|x_n - x_{n-1}|$ also forces convergence, whereas $|x_{n+1} - x_n| \rightarrow 0$ does not.

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1. Prove the monotonicity of the power mean in p , deduce the chain $M_{-1} \leq M_0 \leq M_1 \leq M_2$ together with the cases of equality, and then prove that $M_p \rightarrow \max a_i$ as $p \rightarrow +\infty$.
 2. Prove Toeplitz's lemma and Kronecker's lemma; deduce from the first the theorems on arithmetic and geometric means, and from the second the deterministic form of the strong law of large numbers: if $\sum \frac{x_n}{n}$ converges, then $\frac{x_1 + \dots + x_n}{n} \rightarrow 0$.
 3. Prove both parts of Theorem 2 and construct a sequence with $|x_{n+1} - x_n| \rightarrow 0$ whose set of partial limits is the whole interval $[0, 1]$.
 4. (VJIMC 2011, cat. I) Let $(a_n)_{n \geq 1}$ be an unbounded strictly increasing sequence of positive reals such that the arithmetic mean of any four consecutive terms $a_n, a_{n+1}, a_{n+2}, a_{n+3}$ again belongs to the sequence. Prove that $\frac{a_{n+1}}{a_n}$ converges and find all possible values of the limit.

5. Let $a_n > 0$ satisfy $a_{n+1} \leq a_n + \frac{1}{n^2}$ for all n . Prove that (a_n) converges provided it is bounded below, and show by an example that replacing $\frac{1}{n^2}$ by $\frac{1}{n}$ destroys the statement.